

NUCLEAR POWER REACTORS

Small Modular Reactors

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There is strong interest in smaller units for generating electricity from nuclear power, and for process heat.

Small Modular Reactors (SMRs) represent a broad suite of smaller-scale designs that seek to apply the principles of modularity, factory fabrication, and serial production to nuclear energy.

SMRs offer additional flexibility in operation and wider deployment opportunities, allowing for nuclear to be used in more locations and for a greater range of applications.

What is an SMR?

Small modular reactors (SMRs) are defined as nuclear reactors generally 300 MWe equivalent or less, designed with modular technology using module factory fabrication, pursuing economies of series production and short construction times.

A few small reactors are already in operation and there are dozens of designs in development.

Size

Traditional nuclear reactors produce around 1000 MWe or more of electricity per unit. SMRs, by contrast, are defined by their modest electrical output – typically less than 300 MWe. Some definitions extend to medium-sized reactors of up to 600 MWe. The smallest designs, with capacities typically less than 20 MWe, are sometimes called microreactors.

The smaller capacity of SMRs allows for deployment in settings where large plants may not be practical – such as remote communities, industrial clusters, or regions with small electricity grids.

Modularity

The “modular” in small modular reactor refers to both the design and construction approach.

Both large and small nuclear reactors increasingly make use of modular construction techniques – assembling major components in factories or specialized facilities before transporting them to site. This approach improves quality control, shortens on-site construction times, and can reduce overall costs.

However, SMRs take modularity several steps further:

Factory fabrication: components or even entire reactor modules are designed to be built in factories under controlled conditions.

Serial production: SMR designers plan for serial production to achieve economies of series, similar to those achieved in the aerospace industry.

On-site assembly: factory-fabricated modules are shipped to the site for assembly potentially reducing construction times and costs.

Scalability: many SMRs are designed for incremental deployment, reducing financial risk and providing flexible solutions for customers and end users.

Reactor Technology

SMRs encompass a range of reactor types, many of which are evolutions of existing designs rather than entirely new technologies.

Most SMRs that are candidates for near-term deployment are based on proven water-cooled reactor technologies.

Four main technology options are being pursued: light water reactors, fast neutron reactors, graphite-moderated high temperature reactors and various kinds of molten salt reactors (MSRs). The first has the lowest technological risk, but the second (FNR) can be smaller, simpler and with longer operation before refuelling. Some MSRs are fast-spectrum.

Most SMRs are smaller, simplified LWRs and use the same type of low-enriched uranium fuel with water as coolant. However, some are fast reactors cooled by liquid metals such as sodium or lead. There are also high temperature gas-cooled designs and molten salt reactors in development. Some designs use advanced fuels with higher (5-20%) levels of enrichment (High Assay, Low Enriched Uranium, HALEU) or mixed oxide (MOX) fuel which means they can recycle some materials usually considered waste.

Drivers for SMRs

In the 1950s and 1960s nuclear power plants used reactors that now would be considered small. At that time, many diverse reactor designs were developed, with some being successfully demonstrated. However, from the 1970s onwards utilities increasingly settled on light-water cooled reactors (LWRs) and asked technology vendors to make them larger and larger to achieve economies of scale. Hundreds of units were built and these large units remain the mainstay of global nuclear power.

The current push for SMRs represents rebalancing of historic economies of scale towards economies of series production, as well as a wish to use nuclear energy for more applications than utility scale electricity generation.

Large-scale reactors provide reliable, 24/7 electricity supply on a low cost-per-unit basis, ideal for most on-grid applications. High upfront capital costs are followed by exceptionally low and stable operational costs, meaning it generally makes sense to operate large reactors continuously. Across the range of SMRs in development, new technical features coupled with smaller scale offer potential benefits, particularly in certain deployment scenarios:



Affordability: by virtue of their smaller size, SMRs have a significantly lower capital outlay per unit than large-scale equivalents. This reduces financial risk and allows for a wider range of investors and owners of SMRs.

Modular design: some SMRs are designed to be deployed in modules, allowing capacity to be scaled over time to match demand.

Operational flexibility: modern large-scale reactors can load-follow but are generally operated 24/7. Some SMRs are designed specifically with operational flexibility in mind, with, for example, the ability to quickly switch between electricity and heat generation.

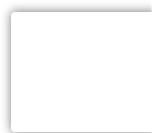
Deployment flexibility: SMRs expand the range of locations and deployment scenarios for nuclear, extending the benefits of nuclear to more end users. SMRs can be deployed on grids that would be too small to accommodate large reactors, whilst smaller core sizes and the passive safety features of some SMRs offer the promise of smaller emergency planning zones, allowing for deployment nearer urban areas and on industrial sites.

SMR development is proceeding in Western countries with a lot of private investment, including small companies. The involvement of these new investors indicates a profound shift taking place from government-led and -funded nuclear R&D to that led by the private sector and people with strong entrepreneurial goals, often linked to a social purpose. That purpose is often deployment of affordable clean energy, without carbon dioxide emissions.

SMR designs and projects

There are a range of over 100 SMR designs at various stages of development. For more information on SMR designs, please visit our [Small Modular Reactor Design Database](#).

There are a small number of SMRs already operating and under construction, as well as numerous projects at various stages of



deployment. For more information on deployment of SMRs, please visit: [Small Modular Reactor \(SMR\) Global Tracker](#).

Fuel

The current fleet of nuclear reactors runs primarily on uranium fuel enriched up to 5% uranium-235 (U-235). High-assay low-enriched uranium (HALEU) is defined as uranium enriched to greater than 5% and less than 20% of the U-235 isotope. Applications for HALEU are today limited to research reactors and medical isotope production. However, HALEU will be needed for many advanced power reactor fuels, and more than half of the small modular reactor (SMR) designs in development.

HALEU is not yet widely available commercially. At present only Russia and China have the infrastructure to produce HALEU at scale. Centrus Energy, in the United States, began producing HALEU from a demonstration-scale cascade in October 2023.

HALEU can be produced with existing centrifuge technology but requires a specific nuclear fuel cycle infrastructure and the development of new or modified regulations and licensing regimes. Moreover, new or modified transport containers will be required for the movement of the large quantities of HALEU required for the deployment of SMRs and advanced reactors.

Establishing the supply chain to produce and deliver HALEU to customers will require significant capital investment. Governments will need to play a role initially until demand from the commercial market provides a sufficient signal to support private investment.

For more information see page on: [High-Assay Low-Enriched Uranium \(HALEU\)](#).

Notes & references

General sources

[Small Modular Reactors Catalogue 2024](#), International Atomic Energy Agency (October 2024)

The NEA Small Modular Reactor Dashboard: Third Edition (September 2025)

Appendix

Military developments of small power reactors from 1950s

US experience and plans

About five decades ago the US Army built eight reactors, five of them portable or mobile. PM1 successfully powered a remote air/missile defence radar station on a mountain top near Sundance, Wyoming for six years to 1968, providing 1 MWe. At Camp Century in northern Greenland the 10 MWt, 1.56 MWe plus 1.05 GJ/hr PM-2A was assembled from prefabricated components, and ran from 1960-64 on high-enriched uranium fuel. Another was the 9 MWt, 1.5 MWe (net) PM-3A reactor which operated at McMurdo Sound in Antarctica from 1962-72, generating a total of 78 million kWh and providing heat. It used high-enriched uranium fuel and was refuelled once, in 1970. MH-1A was the first floating nuclear power plant operating in the Panama Canal Zone from 1968-77 on a converted Liberty ship. It had a 45 MWt/10 MWe (net) single-loop PWR which used low-enriched uranium (4-7%). It used 541 kg of U-235 over ten years and provided power for nine years at 54% capacity factor.

ML-1 was a smaller and more innovative 0.3 MWe mobile power plant with a water-moderated HTR using pressurized nitrogen at 650°C to drive a Brayton closed cycle gas turbine. It used HEU in a cluster of 19 pins, the core being 56 cm high and 56 cm diameter. It was tested over 1962-66 in Idaho. It was about the size of a standard shipping container and was truck-mobile and air-transportable, with 12-hour set-up. The control unit was separate, to be located 150 m away.

All these were outcomes of the Army Nuclear Power Program (ANPP) for small reactor development – 0.1 to 40 MWe – which ran from 1954-77. ANPP became the Army Reactor Office (ARO) in 1992. More recently (2010) the DEER (Deployable Electric Energy Reactor) was being commercialized by Radix Power & Energy for forward military bases or remote mining sites.

A 2018 report from the US Army analysed the potential benefits and challenges of mobile nuclear power plants (MNPPs) with very small modular reactor (vSMR) technology. This followed a 2016 report on Energy Systems for Forward/Remote Operating Bases. The purpose is to reduce supply vulnerabilities and operating costs while providing a sustainable option for reducing petroleum demand and consequent vulnerability. MNPPs would be portable by truck or large aircraft and if abroad, returned to the USA for refuelling after 10-20 years. They would load-follow and run on low-enriched uranium (<20%), probably as TRISO (tristructural-isotropic) fuel in high-temperature gas-cooled reactors (HTRs).

In January 2019 the Department of Defense (DOD) Strategic Capabilities Office solicited proposals for a 'small mobile reactor' design which could address electrical power needs in rapid response scenarios – Project Pele. These would make domestic infrastructure resilient to an electrical grid attack and change the logistics of forward operating bases, both by making more energy available and by simplifying fuel logistics needed to support existing, mostly diesel-powered, generators. They would also enable a more rapid response during humanitarian assistance and disaster relief operations. "Small mobile nuclear reactors have the potential to be an across-the-board strategic game changer for the DOD by saving lives, saving money, and giving soldiers in the field a prime power source with increased flexibility and functionality." The reactors need to be designed to be operated by a crew of six, with one fully qualified engineer and a single operator on duty at all times.

Each reactor should be an HTR with high-assay low-enriched uranium (HALEU) TRISO fuel and produce a threshold power of 1-10 MWe for at least three years without refuelling. It must weigh less than 40 tonnes and be sized for transportability by truck, ship, and C-17 aircraft. Designs must be "inherently safe", ensuring that a meltdown is "physically impossible" in various complete failure scenarios such as loss of power or cooling, and must use ambient air as their ultimate heat sink, as well as being capable of passive cooling. The reactor must be capable of being installed to the point of "adding heat" within 72 hours and of completing a planned shutdown, cool down, disconnect and removal of transport in under seven days. The DOD announced its preparation of an environmental impact statement for the reactor in March 2020, and awarded \$12-14 million contracts to three companies for initial design work. Then BWXT Advanced Technologies and X-energy were selected in March 2021 to develop a final engineering design

by March 2022. Westinghouse has dropped out, and one of the two companies may be commissioned in 2022 to build a prototype reactor.

The DOD in March 2021 said Project Pele is on track for full power testing of a mobile reactor in 2023, with outdoor mobile testing of a prototype microreactor built at Idaho National Laboratory or Oak Ridge National Laboratory in 2024. The programme is also intended to spur commercial development of HTRs. In September 2021 the DOD issued a draft environmental impact statement for the construction and demonstration operation of a prototype mobile microreactor.

In October the US Air Force announced that its first microreactor would be at Eielson air force base in Alaska, near Fairbanks, to be operational in 2027. This does not appear to be part of Project Pele. The base has its own 15 MWe coal-fired power station already, with a railway to supply it with fuel.

Russian experience

The Joint Institute for Power Engineering and Nuclear Research (Sosny) in Belarus built two Pamir-630D truck-mounted small air-cooled nuclear reactors in 1976, during the Soviet era. The entire plant required several trucks. This was a 5 MWt/0.6 MWe HTR reactor using 45% enriched fuel with zirconium hydride moderator and driving a gas turbine with dinitrogen tetroxide through the Brayton cycle. After some operational experience the Pamir project was scrapped in 1985-86. It had been preceded by the 1.5 MWe TES-3, a PWR mounted on four heavy tank chassis, each self-propelled, with the modules (reactor, steam generator, turbine, control) coupled onsite. The prototype started up in 1961 at Obninsk, operated to 1965, and was abandoned in 1969.

Since 2010 Sosny has been involved with Luch Scientific Production Association (SRI SIA Luch) and Russia's N.A. Dollezhal Research and Development Institute of Power Engineering (NIKIET or RDIPE) to design a small transportable nuclear reactor. The new design will be an HTR concept similar to Pamir but about 2.5 MWe.

A small Russian HTR which was being developed by NIKIET is the Modular Transportable Small Power Nuclear Reactor (MTSPNR) for heat and electricity supply of remote regions. It is described as a single circuit air-cooled HTR with closed cycle gas turbine. It uses 20% enriched fuel and is designed to run for 25 years without refuelling. A twin unit plant delivers 2 MWe and/or 8 GJ/h. It is also

known as GREM. No recent information is available, but an antecedent is the Pamir, from Belarus. More recently NIKIET has described the ATGOR – a transportable HTR with up to six parallel commercial gas-turbine engines with two independent heat sources (a nuclear reactor and a start-up diesel fuelled combustor).

Another NIKIET project is the 6 MWt, 1 MWe Vityaz modular integral light water reactor with two turbine generators, which is transportable as four modules of up to 60 tonnes.

In 2015 it was reported that the Russian defence ministry had commissioned the development of small mobile nuclear power plants for military installations in the Arctic. A pilot project being undertaken by Innovation Projects Engineering Company (IPEC) is a mobile low-power nuclear unit to be mounted on a large truck, tracked vehicle or a sledged platform. Production models will need to be capable of being transported by military cargo jets and heavy cargo helicopters, such as the Mil Mi-26. They need to be fully autonomous and designed for years-long operation without refuelling, with a small number of personnel, and remote control centre. It is assumed but not confirmed that these reactors will be the MTSPNR.

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GENERAL ENQUIRIES

t: +44 (0)20 7451 1520
f: +44 (0)20 7839 1501
e: info@world-nuclear.org

MEDIA ENQUIRIES

Henry Preston
t: +44 (0)20 7451 1523
e: press@world-nuclear.org

ADDRESS

York House,
23 Kingsway,
London,
WC2B 6UJ,
United Kingdom

MEMBERSHIP ENQUIRIES

Member support
members@world-nuclear.org

Joining enquiries
enrolment@world-nuclear.org

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